



Effects of substitutional Fe and oxygen vacancies on room temperature ferromagnetism of Fe-doped In_2O_3 dilute magnetic nanowires

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ABSTRACT

The magnetic behavior and origin of dilute magnetic nanowires are critical for further development and application of nanoscale spintronic devices. In this work, the Fe doping into single-crystalline In_2O_3 nanowires are fabricated using chemical vapor deposition (CVD) method to research these problems based on theoretical and experimental analysis. The XRD, XPS, TEM and XAS results show that the deposited Fe-doped In_2O_3 nanowires possess the cubic bixbyite structure and the doped Fe ions substitute for the In sites of In_2O_3 lattice with +2 and +3 mixed valences. The PL results prove the existence of single ionized oxygen vacancies as well as the defect complexes of donor (oxygen vacancy)-acceptor (Fe impurity) pairs. The intrinsic room temperature ferromagnetism can be observed for the Fe-doped In_2O_3 nanowires and the saturation magnetization largely enhance with Fe doping. The first principles calculations show that the introducing of oxygen vacancy defects can cause strong hybridization between Fe 3d and O 2p orbitals at the Fermi level, resulting in the transition from antiferromagnetic to ferromagnetic coupling between the Fe ions. These results open up the possibilities for exploring spin-electronic devices.

1. Introduction

Diluted magnetic semiconductor (DMS) refers to a magnetic semiconductor in which a small amount of non-magnetic metal cations in traditional semiconductors then replaced by 3d transition metals (TM) or 4f rare-earth metals, which exhibit excellent electrical and magnetic properties and become the ideal materials for spintronic devices [1,2], such as spin field effect tubes, spin light-emitting diodes and spin quantum computers. Since Dietl et al. theoretically predicted that room temperature (RT) ferromagnetism can be achieved by doping TM atoms into ZnO and GaN [3], the wide band gaps semiconductors receive a great deal of interest. Especially, the TM-doped TiO_2 , SnO_2 and In_2O_3 have been reported to exhibit high-temperature ferromagnetism [4,5]. Indium oxide (In_2O_3) is a wide band gap n-type semiconductor with a direct band gap of 3.75 eV and bixbyite structure [6]. Due to its excellent optical, chemical, and electronic properties, TM-doped In_2O_3 has become an ideal DMS material for achieving RT ferromagnetic ordering. Although RT ferromagnetism observed in some Fe, Co, and Mn doped In_2O_3 systems is often reported [7–9], the magnetic results from different research groups still raise doubts due to the strong debate for

the magnetic mechanisms and origin. As is well known, the deposition temperature, content of TM ions and oxygen pressure strongly influence the magnetic properties of TM-doped In_2O_3 systems. Especially, the high content of TM doping can result in forming the secondary magnetic phases, including TM metal and oxide clustering, which are strong correlation with magnetic properties observed in the studied systems. For examples, Li et al. proved that the oxygen vacancy defects play an important role on tuning ferromagnetic ordering for the deposited Fe/Sn-codoped In_2O_3 films [10]. He et al. deposited Fe and Cu co-doped In_2O_3 films, and concluded that the observed ferromagnetism is intrinsic and does not derive from TM impurities [11]. However, Jiang et al. reported that the metallic Fe clusters are responsible for the observed ferromagnetism for the Fe/Cu codoped In_2O_3 films [12]. The high solid solubility of Fe element up to 20% in the In_2O_3 matrix has attracted considerable interest. RT ferromagnetism after Fe doping has been observed for the flake [13,14], powder [15] and bulk In_2O_3 [10]. However, limited reports are available in the Fe-doped In_2O_3 nanowires, which possess a large surface-to-volume ratio and broken symmetry. In this letter, the chemical vapor deposition (CVD) method is used to deposit the Fe-doped In_2O_3 nanowires, and the local Fe structure, magnetic

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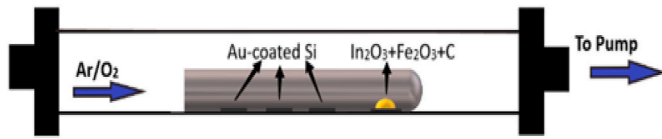


Fig. 1. Schematic diagram of the CVD setup for the preparation of Fe-doped In_2O_3 nanowires.

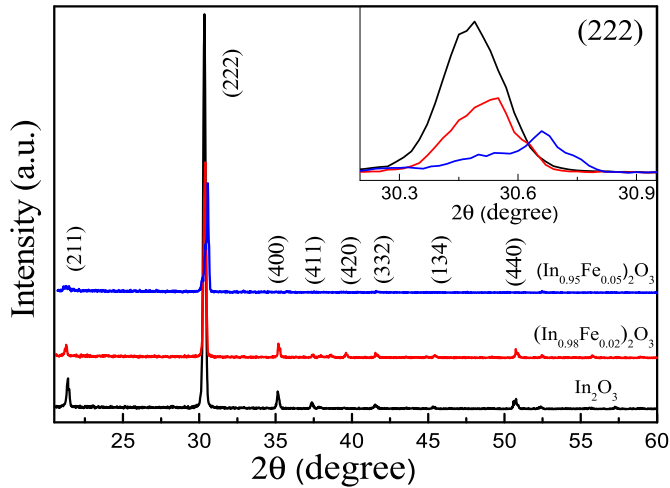


Fig. 2. XRD patterns of undoped, 2% and 5% Fe-doped In_2O_3 nanowires. The inset is the magnification of (222) diffraction peaks.

and optical properties are systematically investigated using XRD, SEM, TEM, XPS, XAFS and PL techniques, which can help to obtain the occupation sites and valence state of Fe dopant, and understand the relationship of ferromagnetic ordering, Fe ions and oxygen vacancies in the Fe-doped In_2O_3 nanowires.

2. Experimental details

The Fe-doped In_2O_3 nanowires are prepared by CVD method. The surface of Si (100) substrates is sprayed with a thin film of Au (about 10 nm). In detail, the one-end sealed quartz tube is filled with the high purity Fe_2O_3 (99.99%), In_2O_3 (99.99%) and C (99.99%) mixed powders as well as the Si substrates. By adjusting the molar ratios between Fe_2O_3 and In_2O_3 powders, the 2% and 5% Fe-doped In_2O_3 nanowires can be fabricated. The distance between the mixed powder and Si substrates is 3–5 cm. The one-end sealed quartz installed with mixed powder along with the Si substrates is then moved into a big quartz tube of a furnace, as shown in Fig. 1. Before heating the tube, the quartz tube is introduced with high-purity Ar (99.9999%) with a flow rate of 300 sccm for 40 min in order to purge the air inside. After the purification, the reaction is then carried out at 1000 °C by controlling the Ar flow rate of 60 sccm and O_2 flow rate of 10 sccm. Under the protection of Ar gas, the deposited nanowires are naturally cooled to RT after the reaction. The Rigaku Dmax-2500 X-ray diffractometer with a $\text{Cu K}\alpha$ radiation is used to detect the crystal structure of Fe-doped In_2O_3 nanowires. The field scanning electron microscopy (SEM, JOEL 6700F) and high-resolution transmission electron microscope (TEM, JEOL-2010) are used to determine the morphology and crystal quality of Fe doped In_2O_3 nanowires. The energy dispersive X-ray spectroscopy (EDS) and the X-ray photoelectron spectroscopy (XPS, PHI-1600 photoelectric spectrometer) with a monochromatic $\text{Mg K}\alpha$ X-rays ($h\nu = 1253.6\text{ eV}$) are used for examined

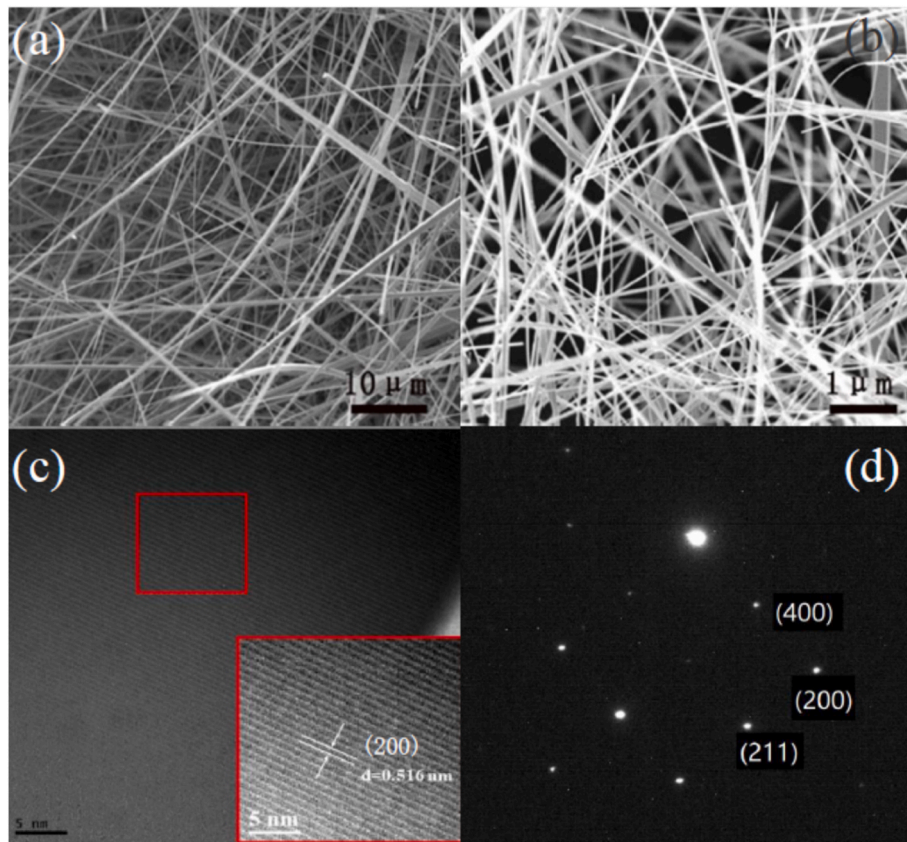


Fig. 3. (a–b) The SEM images of 2% Fe-doped In_2O_3 nanowire with different magnification. (c) The High-resolution TEM image and (d) the corresponding SAED pattern for the 2% Fe-doped In_2O_3 nanowire.

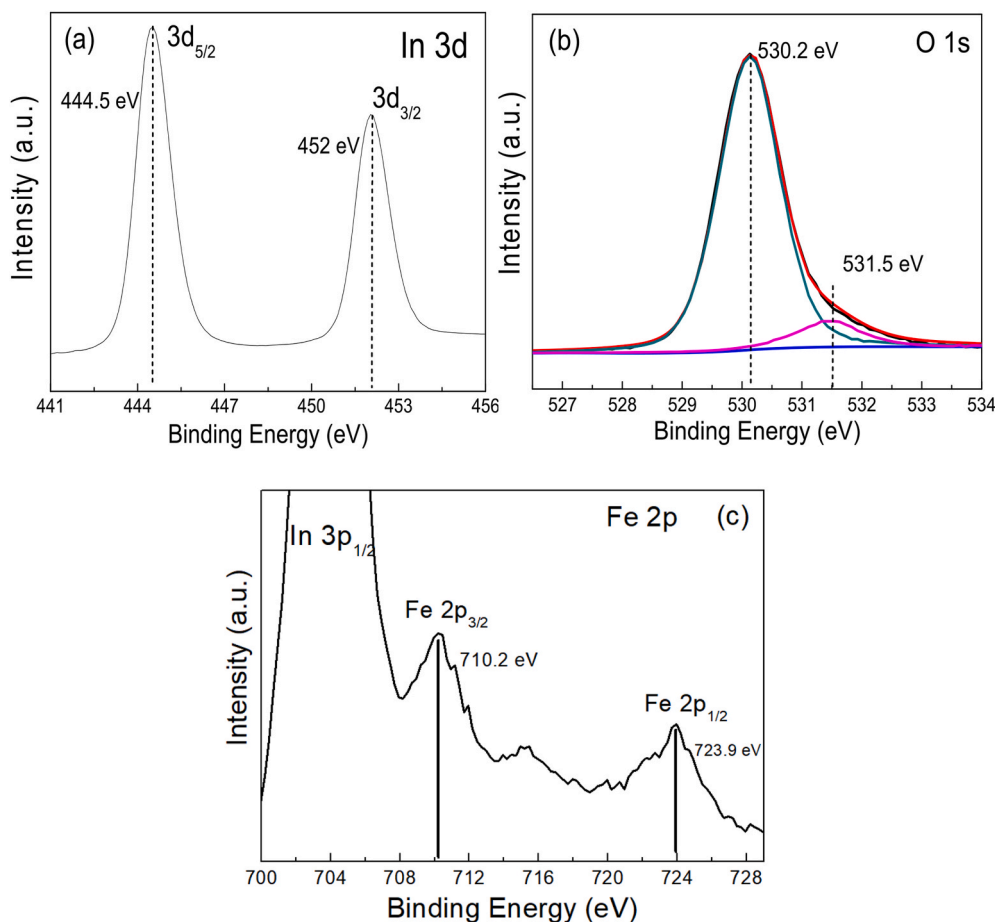


Fig. 4. (a) In 3d, (b) O 1s and (c) Fe 2p XPS spectra of 2% Fe-doped In_2O_3 nanowire.

the composition and valence-state. All the Fe doped In_2O_3 nanowires are engraved by Ar^+ ion sputtering before the XPS measurement in order to remove the surface contamination. The X-ray absorption spectra (XAS) along with X-ray absorption near edge structure (XANES) as well as the extended X-ray absorption fine structure (EXAFS) are measured using the fluorescence mode at 4B9A beam line of X-ray diffraction (XRD) station in Beijing Synchrotron Radiation Facility (BSRF).

3. Results and discussion

Fig. 2 shows the XRD patterns of undoped, 2% and 5% Fe-doped In_2O_3 nanowires. It is obvious that all the samples exhibit a preferred (222) orientation, and the diffraction peaks corresponding to (211), (200), (400), (411), (420), (332), (134) and (440) planes can be indexed to the single-phase cubic bixbyite In_2O_3 structure. The sharp peak suggests a good crystallinity for the deposited nanowires. In order to understand the effects of Fe doping, a careful analysis of the diffraction peaks is carried out. One can see from the inset of Fig. 2 that the (222) diffraction peaks gradually shift to a higher angle from 30.47° to 30.66° with the increase of Fe concentration, implying a gradual decrease in the lattice parameter of In_2O_3 . It is possible that the ion radius of doped Fe element ($R_{\text{Fe}^{3+}} = 0.645 \text{ \AA}$) is smaller than that of In^{3+} ($R_{\text{In}^{3+}} = 0.94 \text{ \AA}$) [16]. The smaller substitutional ion radius will cause lattice shrinkage, and eventually lead to smaller lattice parameter, reflecting that the doped Fe ions are incorporated into the In sites of In_2O_3 lattice.

In order to obtain the morphological characteristics of deposited Fe-doped In_2O_3 nanowires, the SEM technique is employed. Fig. 3(a)–(b) shows the typical SEM images for the Fe-doped In_2O_3 nanowires with different magnification, respectively, which are composed of large amounts of high-quality long nanowires. It can be seen that the whole

length of deposited nanowire reaches micron level (about $10 \mu\text{m}$ – $100 \mu\text{m}$) and the diameters is about 50 – 200 nm . The Fe-doped In_2O_3 nanowires grow disorderly with homogeneous thickness and no impurities and other morphologies are observed. These characteristics indicate that the Fe-doped In_2O_3 nanowires with high quality have been successfully synthesized. The high-resolution TEM images are shown in Fig. 3(c), which reflects the single crystalline nature of the nanowires. The observed interplanar distances for the Fe-doped In_2O_3 nanowires are 0.516 nm , which corresponds to the (200) planes. The selected area electron diffraction (SAED) patterns in the Fig. 3(d) further confirm that the doped In_2O_3 nanowires can be indexed as the expected single crystalline cubic bixbyite In_2O_3 structures without any other secondary phases.

Fig. 4 shows the In 3d, O 1s and Fe 2p core level XPS spectra of 2% Fe-doped In_2O_3 nanowire. Characteristic In $3d_{5/2}$ and In $3d_{3/2}$ splitting due to the spin-orbit coupling can be observed in the XPS spectrum of In 3d state, which is focused in the stronger In $3d_{5/2}$ contribution, as shown in Fig. 4(a). The observed In $3d_{5/2}$ and In $3d_{3/2}$ peaks are located at 444.5 and 452 eV , respectively, which corresponds to the binding energy of In^{3+} in the In_2O_3 lattices and is also consistent with In $3d_{5/2}$ and In $3d_{3/2}$ recorded in the indium oxide standard of bcc structure [17]. Fig. 4(b) shows the O 1s core level XPS spectrum. The O 1s XPS spectrum is asymmetric and exhibits two characteristic peaks, I and II, indicating that there exist two kind of oxygen ions with different chemical environments for the deposited nanowires. The stronger peak I, located at 530.2 eV can be ascribed to O^{2-} ions surrounded by In^{3+} ions in the cubic bixbyite In_2O_3 . The peak II at 531.5 eV is associated with the oxygen ions in the oxygen vacancy (V_O) region in the In_2O_3 matrix [18], indicating that a large amount of V_O is introduced into the doped In_2O_3 nanowires. The Fe 2p XPS spectrum shows two characteristic peaks

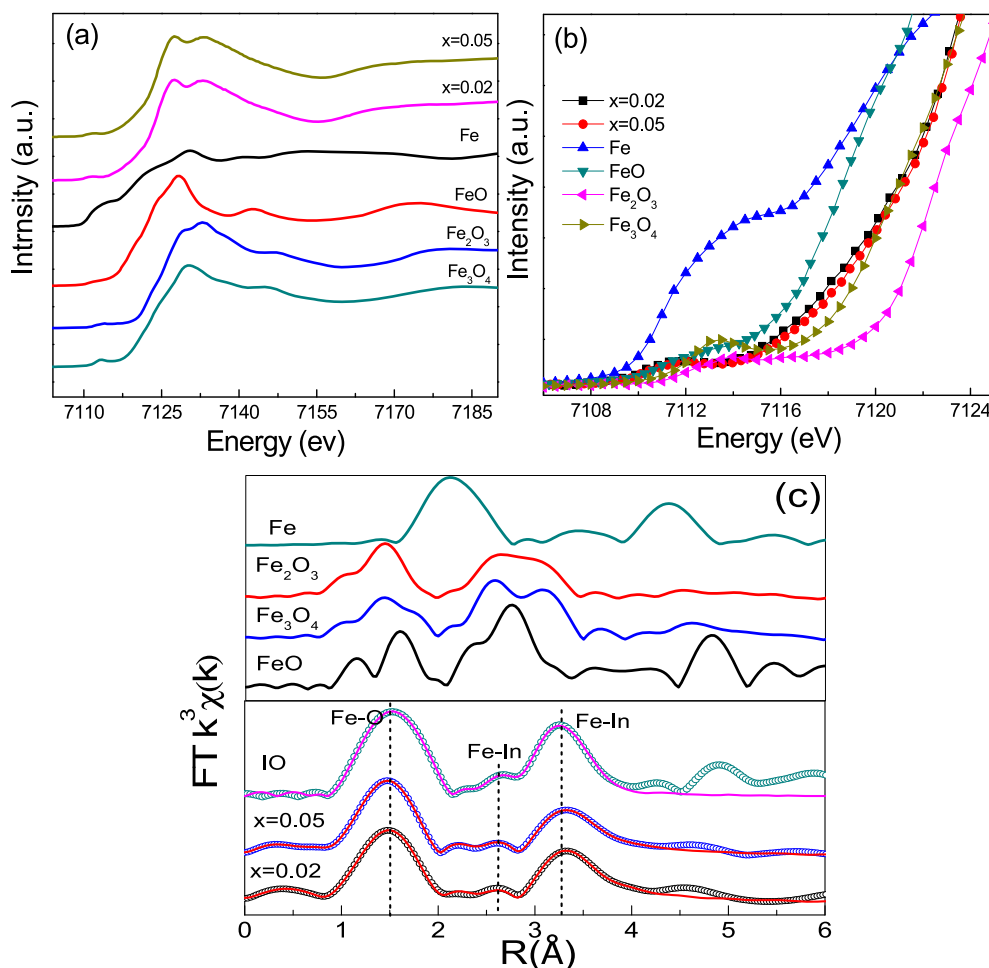


Fig. 5. (a) Fe K-edge XANES spectra and (b) enlarged views of pre-edge absorption features for the 2% and 5% Fe-doped In_2O_3 nanowires as well as Fe, FeO, Fe_2O_3 and Fe_3O_4 foil. (c) Magnitude of Fourier-transformed Fe K-edge and In K-edge EXAFS $k^3\chi(k)$ for the undoped, 2% and 5% Fe-doped In_2O_3 nanowires as well as Fe, FeO, Fe_2O_3 and Fe_3O_4 foil. Dash line, fitting; solid lines, experimental.

around 710.2 and 723.9 eV, which are assigned to the $2p_{3/2}$ and $2p_{1/2}$ states, respectively, as shown in Fig. 4(c). The $2p_{3/2}$ binding energies of Fe^{2+} and Fe^{3+} ions are located at 709.9 eV and 711 eV, respectively, indicating that the doped Fe ions own a mixed-valence of +2 and +3 in the nanowires.

The XANES and EXAFS as spectroscopic tools are a unique local probe technique, which is sensitive to chemical states and local structural properties around TM dopants in the host compounds. Fig. 5(a) shows the Fe K-edge XANES spectra of 2% and 5% Fe-doped In_2O_3 nanowires, together with reference samples, Fe, FeO, Fe_2O_3 and Fe_3O_4 . One can see that there exist some characteristic peaks in the XANES spectra of doped nanowires. The intensity and position of characteristic peaks have a strong correlation with the coordination and symmetry around Fe dopants. These characteristic peaks are rather different from those of reference samples of Fe, FeO, Fe_2O_3 and Fe_3O_4 , revealing that there exist different local environments around Fe dopants in the experimental samples. In other words, the Fe element in the In_2O_3

nanowires is not in the form of Fe or Fe oxide, which indicates that the doped Fe element is likely to replace the In^{3+} site and then immerse it into the In_2O_3 lattice. The enlarged views of pre-edge absorption features of Fe K-edge XANES spectra for the experimental and reference samples are shown in Fig. 5(b). Clearly, the absorption edge positions of experimental samples are closer to the edge of reference Fe_3O_4 and lie between that of FeO and of Fe_2O_3 , which again proves the existence of $\text{Fe}^{2+}/\text{Fe}^{3+}$, which is consistent with the XPS analysis results. Fig. 5(c) shows the Fourier transforms (FTs) of EXAFS oscillation functions $k^3\chi(k)$, representing the radial distribution function at the Fe K-edge for the 2% and 5% Fe-doped In_2O_3 nanowires. As references, the Fe K-edge FT curves of reference samples of Fe, FeO, Fe_2O_3 and Fe_3O_4 , and the In K-edge result of pure In_2O_3 are also plotted. Three peaks located at 1.50, 2.63, and 3.25 Å can be clearly observed at a longer radial distance, which corresponds to the first oxygen shell (Fe–O) and the second and the third In shells (Fe–In), respectively. The position of the three peaks well matches with those of In atoms in pure In_2O_3 , but is completely

Table 1

Best fit structural parameters around Fe dopant for the 2% and 5% Fe doped In_2O_3 nanowires. N, R and σ^2 are the coordination number, bond length and Debye-Waller factor, respectively.

Samples	Fe–O			Fe–In ₁			Fe–In ₂		
	N ₁	σ_1^2 (Å ²)	R ₁ (Å)	N ₂	σ_2^2 (Å ²)	R ₂ (Å)	N ₃	σ_3^2 (Å ²)	R ₃ (Å)
x = 0.02	5.53	0.0082	2.04	5.53	0.0054	3.354	5.53	0.0069	3.793
x = 0.05	5.48	0.0089	2.03	5.48	0.0063	3.351	5.48	0.0071	3.788

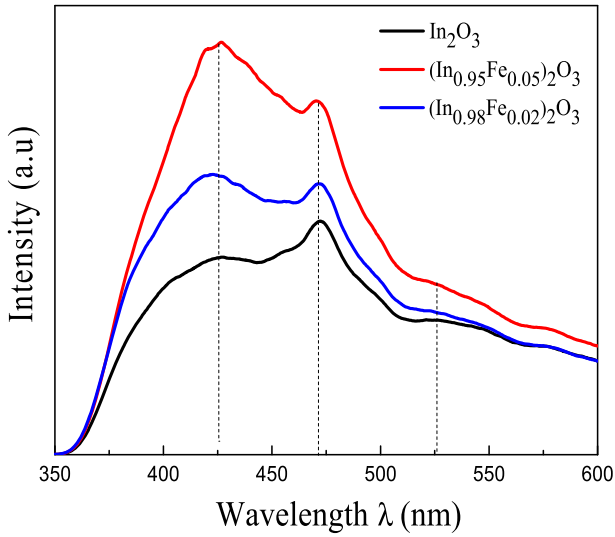


Fig. 6. PL spectra of undoped, 2% and 5% Fe doped In_2O_3 at the excitation wavelength of 320 nm.

different from Fe (2.25, 4.05, and 4.67 Å), FeO (1.25 and 2.73 Å), Fe_2O_3 (1.67, 2.35, and 4.21 Å) and Fe_3O_4 (1.67, 2.35, and 4.21 Å). This suggests that the doped Fe ions are substituted for the In sites of In_2O_3 lattices without forming the Fe metal and Fe oxide secondary phases. In_2O_3 cell is composed of 80 atoms, including 8 In_1 atoms and 24 In_2

atoms, which correspond to trigonally distorted octahedral b sites and tetragonally distorted octahedral d sites, respectively. To obtain quantitative structural information of Fe dopants, the corresponding FT curves of the Fe-K edge EXAFS oscillation function $k^3\chi(k)$ are theoretically fitted by Fe-O and Fe-In coordinations. The theoretical phase shift and backscattering amplitude are calculated from the Fe-substituted In_2O_3 model clusters by the Athena and Artemis interfaces to the IFEFFIT program package [19]. The closest match between the fitted spectra (dash line) and experimental spectra (solid lines) is obtained by assuming Fe atoms substitute for In_1 sites of the In_2O_3 lattice. The best fitted results are shown in Table 1. It is obvious that the Fe-O bond length is smaller than the In-O bond length of In_2O_3 ($R_{\text{In-O}} = 2.19$ Å), which is due to the lattice compression caused by Fe-substitution. The large $(\sigma_{\text{Fe-O}})^2$ value in the first nearest neighbor around the Fe atoms can be attributed to the enhancement of the lattice shrinkage by Fe doping as well as the relaxation of the oxygen environment of Fe atoms upon substitution. These results further suggest that the doped Fe ions substitute for the In sites in the In_2O_3 matrix without forming Fe metal clusters and Fe oxide secondary phases in the deposited In_2O_3 nanowires (see Fig. 6).

The photoluminescence (PL) data of Fe doped In_2O_3 nanowires reveal the relationship between magnetic properties and defects. The PL peaks of doped samples are the same as those of pure In_2O_3 nanowires at 425 nm, 473 nm and 526 nm, as shown in Fig. 6. The similar peak patterns of PL spectra suggest that the crystal structures of Fe doped In_2O_3 nanowires are the same as that of pure In_2O_3 nanowires. The In_2O_3 nanowires exhibit green emission at 546 nm, which may be caused by the transition to electrons from deep donor energy level of oxygen

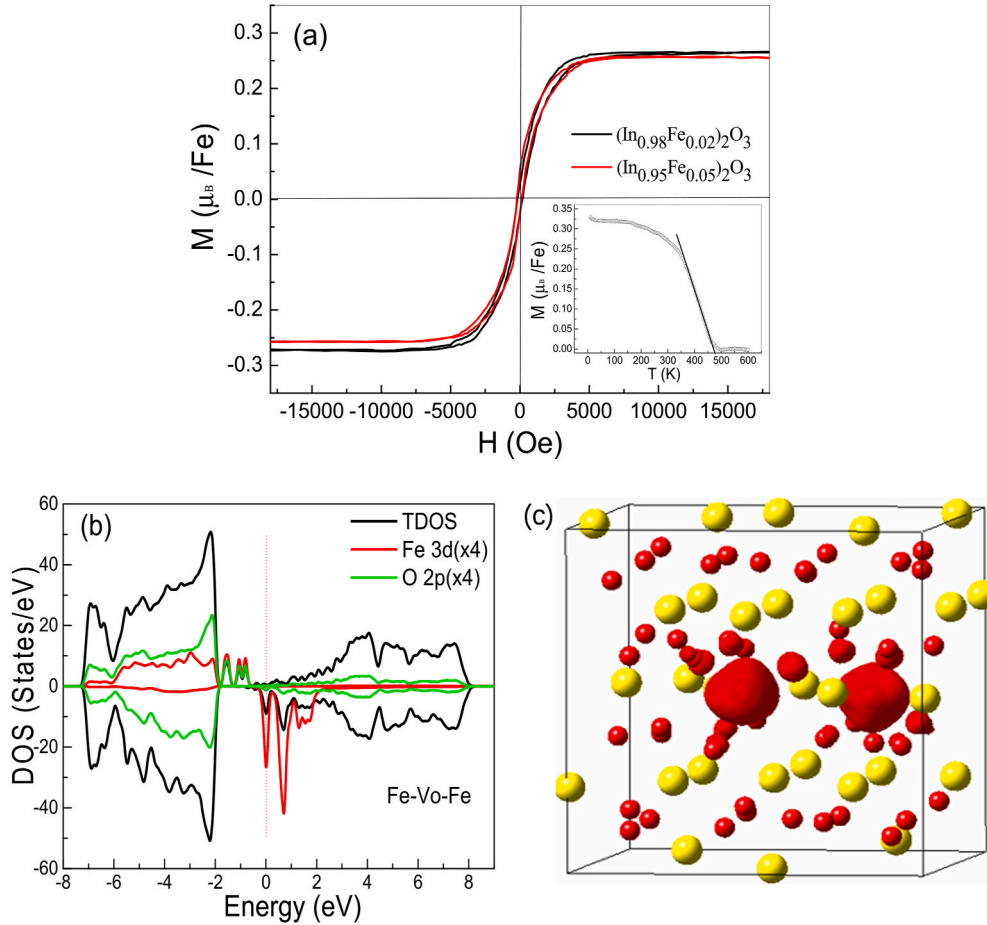


Fig. 7. Magnetic hysteresis loops ($M-H$) for the 2% and 5% Fe-doped In_2O_3 nanowires at 300 K. The inset is temperature dependent magnetization of 2% Fe-doped In_2O_3 nanowires measured under a magnetic field of 50000 Oe. (b) The density of states and (c) spin density distribution in the configuration for the Fe-doped In_2O_3 with oxygen vacancy. Fermi energy level is set at $E_f = 0$ eV.

vacancies (V_O) to valence band [20]. The blue luminescence at 473 nm can be attributed to the recombination of electrons from the single ionized oxygen vacancies (V_O^+) with photo generated holes [21,22]. The ultraviolet luminescence at 425 nm is still in debate. Some reports show that the In_2O_3 nanowires exhibit an ultraviolet luminescence at 422 nm, which originates from the oxygen vacancy defects [23]. The similar ultraviolet luminescence (413 nm) observed in the Cr-doped In_2O_3 nanostructure is attributed to the near band edge emission [24]. For the Fe doped In_2O_3 nanowires, it is obvious that the intensity of ultraviolet emission (422 nm) gradually increases with Fe doping, implying a strong correlation with the Fe concentration. The ultraviolet emission of 422 nm possibly originates from the radiative recombination of donor (oxygen vacancies)-acceptor (Fe ions) pairs. With the increase of Fe concentration, more donor-acceptor pairs are formed, resulting a gradual increase in the intensity of ultraviolet emission, which is consistent with the reported results in the Cu-doped SnO_2 films [25].

Fig. 7(a) shows the RT $M-H$ curves for the 2% and 5% Fe-doped In_2O_3 nanowires. The diamagnetic background signals from the Si substrates have been removed from the raw data. Clearly, the observed $M-H$ curves exhibit well-defined open and symmetric hysteresis loops at 300 K, strongly suggesting that the deposited Fe-doped In_2O_3 nanowires exhibit RT ferromagnetic behavior. The saturation magnetization and coercive field are $0.266 \mu_B/\text{Fe}$ and 145 Oe for the 2% Fe-doped In_2O_3 nanowire. With increasing Fe doping concentration to 5%, the saturation magnetization and coercive field reaches to 0.25 emu/g and 182 Oe. The slight decrease in saturation magnetization with Fe doping can be attributed to an increase in the antiferromagnetic coupling of Fe ions pairs. The inset of Fig. 7(a) shows the temperature dependent magnetization for the 2% Fe-doped In_2O_3 nanowire. Clearly, the magnetization decreases slowly in the temperature range of 10–200 K and decreases rapidly with further increasing temperature. The Curie temperature T_C can be estimated to be as high as 480 K. The XRD, XPS, TEM and EXAFS results have clearly shown that the doped Fe ions replace the In^{3+} site into the In_2O_3 lattices and the Fe metal or Fe oxide second phases can be safely excluded, implying that the observed ferromagnetic ordering is intrinsic. In order to further reveal the mechanisms of observed RT ferromagnetic ordering, the first principles calculations are used to calculate the electronic structure of Fe-doped In_2O_3 by the Vienna *ab initio* simulation package (VASP). According to the XPS and PL results, a large amount of oxygen vacancies is induced in the deposited Fe-doped In_2O_3 nanowires. Therefore, the Fe- V_O -Fe configuration is established, namely, two Fe atoms substitute for the nearest neighboring In atoms and the O atom between Fe atom is removed to introduce an oxygen vacancy. The calculated ferromagnetic stability energy ($\Delta E = E_{\text{FM-AFM}}$) is -22.6 meV by considering the energy difference between ferromagnetic and antiferromagnetic ground states, suggesting that the Fe- V_O -Fe configuration is stable ferromagnetic ground state due to $\Delta E < 0$. Generally, the Fe-O-Fe configuration exhibits a stable antiferromagnetic interaction by the hybridization between the Fe 3d and O 2p states. However, the existence of oxygen vacancy results in the transition from antiferromagnetic to ferromagnetic coupling. The total and partial density of states of Fe- V_O -Fe configuration are shown in Fig. 7(b). One can see that more spin-up Fe 3d electrons are distributed in the valence band, resulting in forming a spin-down Fe 3d minority states in the conduction band. On the other hand, the Fe 3d and O 2p orbitals strongly hybridize around the Fermi level, which causes a large energy splitting and leads to the ferromagnetic coupling between two Fe atoms. Fig. 7(c) shows the spin density distribution of Fe- V_O -Fe configuration. The red isosurfaces correspond to the spin-up region, and the large red balls represent the two Fe atoms. It is obvious that the two Fe atoms have same spin polarization direction, further confirming the existence of ferromagnetic coupling between two Fe atoms. The induced magnetic moment is mainly contributed by the Fe 3d orbital, corresponding to a total magnetic moment of $7.41 \mu_B$. Therefore, it can be concluded that the observed RT ferromagnetic ordering in the Fe-doped In_2O_3 nanowires strongly correlates with the coexistence of Fe doping and oxygen

vacancy defects. The oxygen vacancy V_O can act as a bridge to mediate the indirect ferromagnetic coupling between Fe atoms.

4. Conclusion

In summary, the effects of Fe doping on local structure, optical and magnetic properties of Fe-doped In_2O_3 dilute magnetic nanowires fabricated by CVD method are investigated in detail by XRD, XPS, SEM, TEM, XAS, PL and magnetic measurements. The results show that the deposited Fe-doped In_2O_3 nanowires exhibit a single-phase cubic bixbyite In_2O_3 structure with the substitutional Fe at the In^{3+} sites of In_2O_3 lattices. No extra Fe metal and other Fe oxide impurity phases can be detected. The PL measurements prove the existence of the single ionized oxygen vacancies as well as the Fe- V_O defect complexes, which are strongly correlation with the RT ferromagnetic ordering observed for the Fe-doped In_2O_3 nanowires. Based on the first principles calculations, the observed RT ferromagnetism in the Fe-doped In_2O_3 nanowires originates from the Fe- V_O -Fe superexchange interactions. The introducing of oxygen vacancy defect can cause the strong hybridization between the Fe 3d and O 2p orbitals around the Fermi level, which acts as a bridge to mediate the indirect ferromagnetic coupling between Fe atoms.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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